# Efthymios Costa

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GitHub Profile
Personal Website


## Education

**PhD in Statistics and Machine Learning (StatML)** *Oct 2021 - Sep 2025*
*Imperial College London, Department of Mathematics*
Supervisors: Dr Ioanna Papatsouma & Professor Alastair Young
Thesis title: *Novel unsupervised techniques for mixed-type data*

**Visiting Doctoral Student** *Jan - Mar 2024*
*University of Bologna, Department of Statistical Sciences "Paolo Fortunati"*
Visiting Supervisor: Professor Christian Martin Hennig

**MSc Statistics (Data Science)** *Sep 2020 - Sep 2021*
*Imperial College London, Department of Mathematics* Distinction (80.89/100)
Thesis: "A benchmarking study of distance-based clustering methods for mixed-type data"

**BSc Mathematics** *Sep 2017 - Jun 2020*
*Imperial College London, Department of Mathematics* First Class Honours (78.21/100)

## Employment

**Research Associate with Teaching Duties in Statistics** *Oct 2025 - Present*
*Imperial College London, Department of Mathematics* London, United Kingdom
Postdoc & Fellows Representative in the Department of Mathematics

## Research Interests

Unsupervised learning: Cluster analysis, outlier detection, dimensionality reduction.

Mixed-type data: Combination of continuous and discrete variables in data.

Robust statistics: Robustness guarantees for machine learning methodologies.

## Publications & Preprints

**Efthymios Costa**, Christian Martin Hennig (2026). "A unified approach to outlier identification for mixed-type data" *(manuscript in preparation).*

**Efthymios Costa**, Ioanna Papatsouma, Angelos Markos (2026). "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data". *To appear in Pattern Recognition.*

**Efthymios Costa**, Ioanna Papatsouma, Angelos Markos (2026). "Sparse clustering via the Deterministic Information Bottleneck algorithm". *arXiv preprint arXiv:2601.20628.*

**Efthymios Costa**, Ioanna Papatsouma (2025). "A novel framework for quantifying nominal outlyingness". In: *Statistics and Computing* 36, pp. 41–58.

**Efthymios Costa**, Ioanna Papatsouma (2025) Contribution to the discussion of Dümbgen and Davies (2025) "Connecting Model-Based and Model-Free Approaches to Linear Least Squares Regression". In: *Statistica* 84, pp. 107–108.

**Efthymios Costa**, Ioanna Papatsouma, Angelos Markos (2022). "Benchmarking distance-based partitioning methods for mixed-type data". In: *Advances in Data Analysis and Classification* 17, pp. 701—724.

## Funding & Awards

**Poster Competition 1st Place & People's Choice Award**
*Faculty of Natural Sciences Research Showcase 2025, Imperial College London*

**Poster Competition People's Choice Award**
*Mathematics PhD Symposium 2025, Imperial College London*

**Best Postgraduate/Doctoral Paper Award**
*6th Annual Conference of the Cyprus Statistical Society, 2025*

**Faculty of Natural Sciences Prize for Excellence in Teaching**
*Highly commended nomination for excellence in teaching and learning, 2025, Imperial College London*

**MSc in Statistics Imperial College London Ambassador**
*Honorary title in recognition of outstanding contributions to the running and delivery of the programme*

**Poster Competition Runner-up & People's Choice Award**
*Mathematics PhD Symposium 2022, Imperial College London*

**PhD Studentship**
*EPSRC Centre for Doctoral Training (CDT) in Statistics and Machine Learning (StatML)*

**Warner Prize**
*Prize awarded for research project development in the MSc in Statistics at Imperial College London - Class of 2021*

**Outstanding Academic Representation Network Team Award**
*Student volunteer award - Imperial College Union Student Choice Awards 2020*

## Qualifications

**Associate Fellow of Advance HE (AFHEA)**
*Awarded on February 2024, Imperial College London*

## Conferences, Workshops & Seminars

**19th International Conference on Computational and Methodological Statistics** *13 - 15 Dec 2025*
*Session Title: "Clustering of heterogeneous data"* Session Organiser/Chair

**Joint meeting of the Italian and Dutch/Flemish Classification Societies (CLADAG-VOC 2025)** *8 - 10 Sep 2025*
*Presentation title: "Cluster Analysis from an Information-Theoretic Viewpoint"* Invited Speaker

**International Conference on Robust Statistics (ICORS 2025)** *19 - 23 May 2025*
*Presentation title: "A novel framework for quantifying nominal outlyingness"* Invited Speaker

**18th International Conference on Computational and Methodological Statistics** *14 - 16 Dec 2024*
*Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data"* Presenter

**RSS Emerging Applications Section Meeting "Network and Clustering Analysis for Emerging Applications"** *13 Dec 2024*
*Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data"* Invited Speaker

**UKRI EPSRC Mathematical Sciences Team visit at Imperial College London** *9 Oct 2024*
*Poster title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data"* Presenter

**18th Conference of the International Federation of Classification Societies** *16 Jul 2024*
*Presentation title: "A Deterministic Information Bottleneck Method for Clustering Mixed-Type Data"* Presenter

**Imperial Mathematics PhD Symposium** *12 Jun 2024*
*Poster title: "A Novel Approach to Outlier Detection for Mixed-Type Data"* Presenter

**Department of Statistical Sciences "Paolo Fortunati", University of Bologna - Weekly Statistics Seminars** *1 Feb 2024*
*Presentation title: "Outlier detection for mixed-type data: A novel approach"* Presenter

**16th International Conference on Computational and Methodological Statistics** *16 - 18 Dec 2023*
*Poster title: "A novel approach to outlier detection for mixed-type data"* Presenter

**Amazon Development Center Germany GmbH** *27 - 28 Mar 2023*
*Advanced training on Statistical Machine Learning and Computing for big-data analysis* Attendee
*Poster title: "Benchmarking distance-based partitioning methods for mixed-type data"* Presenter

**Faculty of Natural Sciences Research Showcase 2022** *29 Sep 2022*
*Presentation title: "Clustering mixed-type data: Which method to choose?"* Presenter

**Bocconi Summer School in Advanced Statistics & Probability** *11 - 22 Jul 2022*
*Topic: Random Structures and Combinational Statistics* Attendee

**Imperial Mathematics PhD Symposium** *16 Jun 2022*
*Poster title: "A full factorial benchmarking study of non-parametric partitioning methods for mixed-type data"* Presenter

## Teaching Experience

**BSc/MSci Mathematics Final Year Project Supervisor**
Imperial College London, Department of Mathematics (Academic Year 2025-2026)

**MSc Statistics Dissertation Supervisor**
Imperial College London, Department of Mathematics (Academic Year 2025-2026)

**Senior Graduate Teaching Assistant**
*Probability and Statistics* - BSc/MSci Mathematics Year 1 (Autumn 2024)
*Probability for Statistics* - BSc/MSci Mathematics Year 2 (Autumn 2022)
Imperial College London, Department of Mathematics

*Statistics* - MEng Mechanical Engineering Year 3/4 (Autumn 2023)
Imperial College London, Department of Mechanical Engineering

**Graduate Teaching Assistant**
*Data Science* - MSc Statistics (Spring 2022)
*Computational Statistics* - MSc Statistics (Autumn 2022, 2023 & 2024)
*Unsupervised Learning* - MSc Machine Learning and Data Science (Autumn 2022)
*Deep Learning, Supervised Learning* - MSc Machine Learning and Data Science (Spring 2023)
*Machine Learning* - MSc Statistics (Spring 2023)
*Introduction to Statistical Learning, Statistical Modelling II* - BSc/MSci Mathematics Year 3/4 (Spring 2023)
*Statistical Modelling I* - BSc/MSci Mathematics Year 2 (Spring 2023)
Imperial College London, Department of Mathematics

## Outreach

Mentor for Imperial College London Mathematics School Research Project (2024 - 2025)

## Editorial & Peer-Review Service

**Reviewer**

*Advances in Data Analysis and Classification*
*International Statistical Review*
*Journal of Classification*
*Mathematics*

## Memberships

Fellow of the Royal Statistics Society (RSS)

Member of the British Data Science Society (BDSS)

Member of the Cyprus Statistics Society (CSS)

## Skills

**Programming/IT**: R (Advanced/Preferred), Python (Intermediate), C++ (Basic), MATLAB (Basic), LaTeX, HPC projects on compute clusters, Office 365

**Languages**: Greek (native), English (fluent), French (proficient)